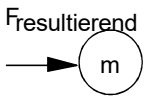
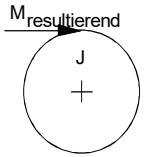
Formelsammlung f. Klausur - Allgemeine Zusammenhänge

$$M_M = M_{Last,MS} + 2\pi \cdot (J_M + J_{Getriebe,MS} + J_{Last,MS}) \cdot dn_M / dt$$

Motorbetrieb, wenn $M_{Motor} \cdot n_{Motor} > 0$





$$v = \frac{ds}{dt} \quad s = \int_0^{t_0} v dt + s_0 \qquad \omega = \frac{d\varphi}{dt} \quad \varphi = \int_0^{t_0} \omega dt + \varphi_0$$

$$a = \frac{dv}{dt} = \frac{d^2 s}{dt^2} \quad v = \int_0^{t_0} a dt + v_0 \qquad \alpha = \frac{d\omega}{dt} = \frac{d^2\varphi}{dt^2} \qquad \omega = \int_0^{t_0} \alpha dt + \omega_0$$

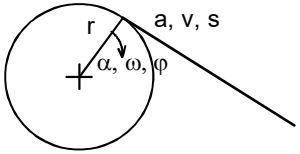
$$F = m \cdot a \qquad M = J \cdot \alpha = J \cdot \frac{d\omega}{dt} = 2\pi \cdot J \cdot \frac{dn}{dt}$$

$$W = \frac{m}{2} v^2 \qquad W = \frac{J}{2} \omega^2$$

$$W = F \cdot s \qquad W = M \cdot \varphi$$

$$P = F \cdot v \qquad P = M \cdot \omega = M \cdot 2\pi \cdot n$$

$$M = F \cdot r$$



$$s = r \cdot \varphi \qquad v = r \cdot \omega \qquad a = r \cdot \alpha \qquad \omega = 2\pi \cdot n$$

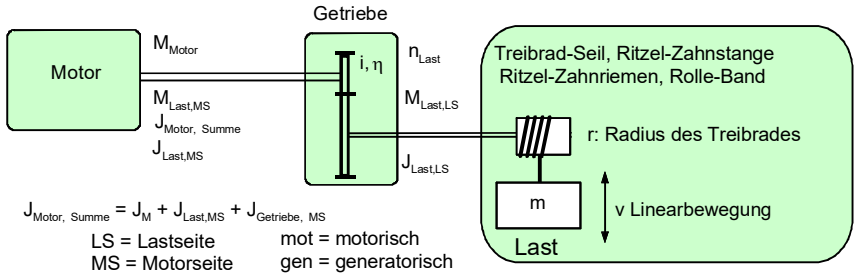
Hochlaufzeiten $\Delta t = \dfrac{2\pi J_{Motor,Summe} \cdot \Delta n}{\overline{M}_M - \overline{M}_{Last}}$

Servomotoren

$$n_{Spiel} = \frac{\sum_{i=1}^{m} \left|\overline{n}_i\right| \cdot \Delta t_i}{\sum_{i=1}^{m} \Delta t_i} \qquad M_{eff} = \sqrt{\frac{\sum_{i=1}^{m} M_{Motor_i}^2 \cdot \Delta t_i}{\sum_{i=1}^{m} \Delta t_i}}$$

Spannungsgrenzkurve, max. Moment, max. Drehzahl

Bewegungen



$J_{Motor,\ Summe} = J_M + J_{Last,MS} + J_{Getriebe,\ MS}$

LS = Lastseite
MS = Motorseite

mot = motorisch
gen = generatorisch

$s = r \cdot \varphi \quad v = r \cdot \omega \quad a = r \cdot \alpha$

Linearbewegung ⇔ Drehbewegung: $J_{Last,LS} = m \cdot r^2$

Drehbewegung ⇔ Drehbewegung $i = \dfrac{n_{Motor}}{n_{Last}} \quad J_{Last,MS,mot} = \dfrac{J_{Last,LS}}{i^2}$

Motorisch: $M_{Last,MS} = \dfrac{M_{Last,LS}}{i \cdot \eta_{Getriebe}}$ Generatorisch:

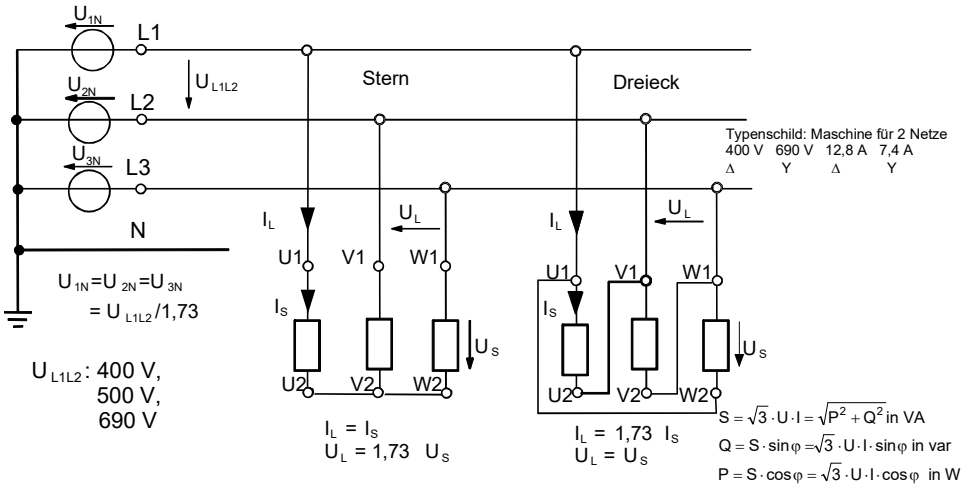
$$M_{Last,MS} = \frac{M_{Last,LS} \cdot \eta_{Getriebe}}{i}$$

Drehstrom

Scheinleistung S: $S = \sqrt{3} \cdot U \cdot I = \sqrt{P^2 + Q^2}$ in VA | $Q = S \cdot \sin\varphi = \sqrt{3} \cdot U \cdot I \cdot \sin\varphi$ in var Blindleistung Q

Wirkleistung P: $P = S \cdot \cos\varphi = \sqrt{3} \cdot U \cdot I \cdot \cos\varphi$ in W

$n_D = f_1/p$



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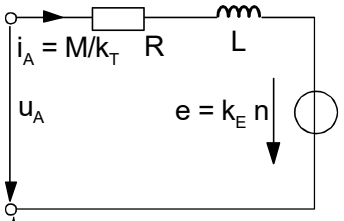
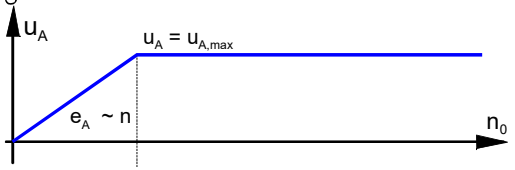
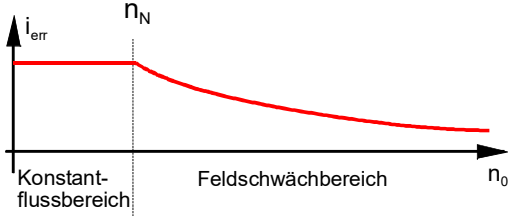
DCM

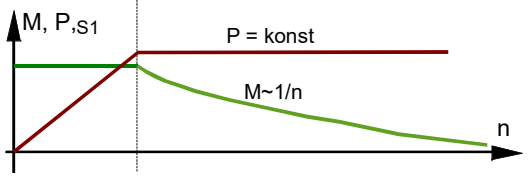
Erregerkreis:

$$u_{err} = i_{err} \cdot R_{err} + L_{err} \frac{di_{err}}{dt}$$









Läuferkreis: $u_A = e_A + i_A \cdot R_A + L_A \frac{di_A}{dt}$

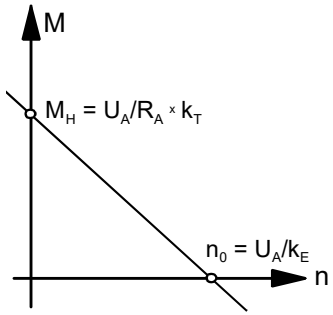
Induzierte Spannung:

$e_A = k_E \cdot n \qquad n = k_n \cdot e_A$

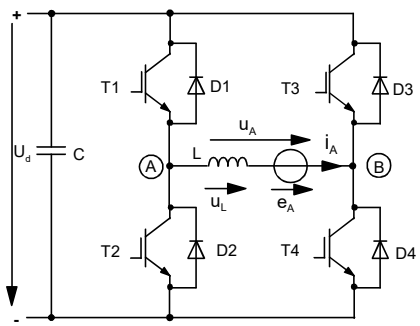
Drehmoment: $M_{innen} = \frac{k_E}{2\pi} \cdot i_A = k_T \cdot i_A$

$$M_{innen} - M_V - M_{Last} = 2\pi \cdot J_{Motor+Last} \cdot \frac{dn}{dt}$$

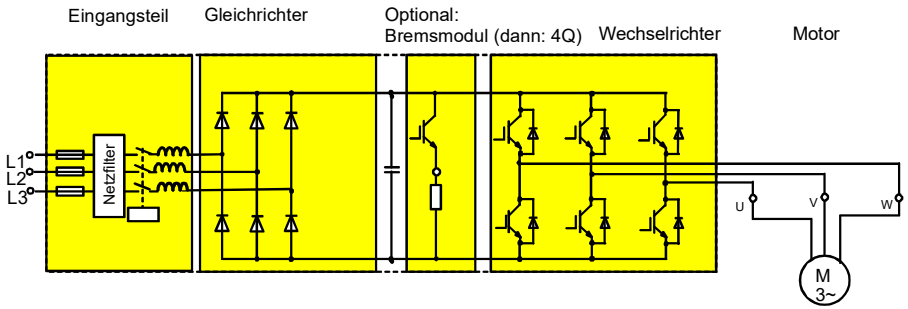
$$n = \frac{u_A}{k_E} - \frac{R_A \cdot 2\pi}{k_E^2} \cdot M$$



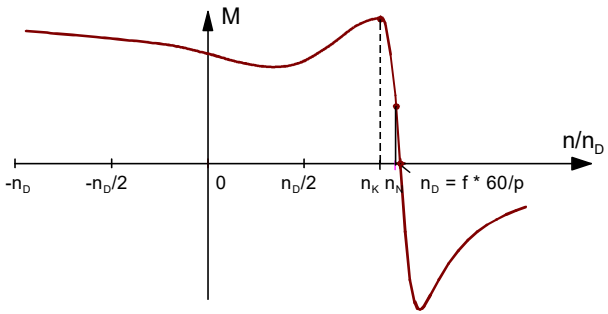
H-Brücke



Umrichter



ASM



Hoher Einschaltstrom!

Drehmoment abhängig vom Quadrat der Spannung

n_N ca. knapp unter f·60 /p

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